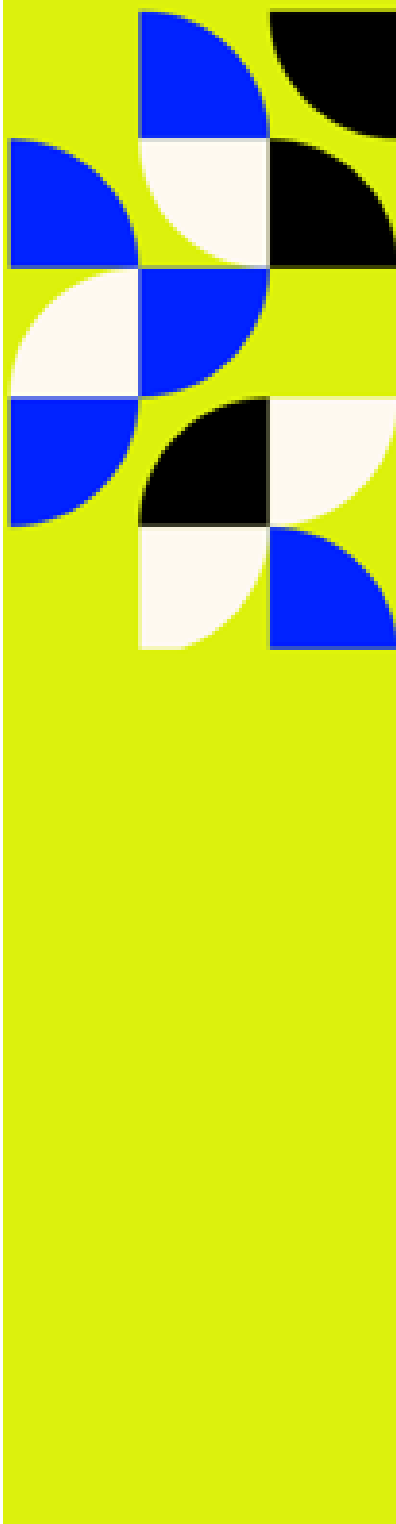




# Breaking Down Yellevate's Customer Dispute Problem

Group 13

Condrada, Jun Frank

Gilles, Joshua Andrei Bon T.

Lim, Gerald M.

Ramos, Bryan

Rangasan, Relie

Reyes, Fritz Gerald

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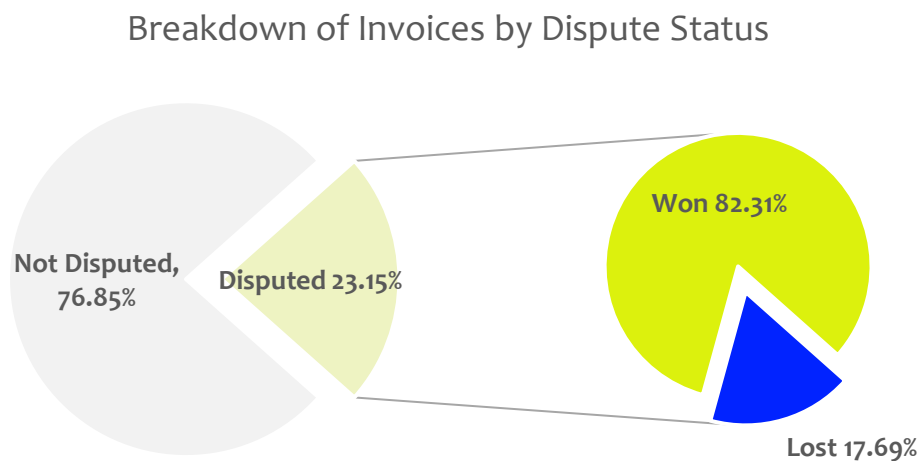
# PROBLEM

## Problem Statement

Our team was approached by Yellevate’s top management to get insights on how to reduce the revenue loss from disputed invoices. For the past few years, the company has been struggling with a surge of clients disputing their invoices. Yellevate defines disputes as clients expressing dissatisfaction with the company’s services and refusing to pay for them. Lost disputes mean that the company does not get paid for the work it has done.

This has been a huge financial burden for the company. Throughout 2020 and 2021, 27.25 percent or 571 of all invoices issued by the business have gone under a dispute process. Of these 571 disputes, 17.69% were lost which translated to a 4.67 percent loss (or \$690,167), compared to the total possible revenue if all invoices were collected. Currently, there are no solutions in place for this issue.

Figure 1. Breakdown of Invoices by Dispute Status



The top management believes that the surge in disputes is caused by clients thinking they can get away with not paying the company. Further, they believe that all the company’s obligations to it have been fulfilled and that the quality of the services rendered is not the cause of the increased disputes.

## Company Profile

Yellevate focuses on helping other businesses advertise themselves. Email marketing, website design, content generation, and other marketing initiatives are among the things that Yellevate assists mid-sized businesses with launching.

Yellevate monitors how successfully it aids businesses in enhancing their marketing operations using data analytics. For instance, Yellevate monitors this by looking at email impressions, click-through rates, website traffic, and other key performance metrics if a business contacts them for assistance in improving its online advertising operations.

## Data Analysis Goals

At the minimum, the team were asked to provide insights for the following:

- a. The processing time in which invoices are settled (average # of days rounded to a whole number)
  - On average, all invoices are processed within 26 days (see Figure 8 on page 18).
- b. The processing time for the company to settle disputes (average # of days rounded to a whole number)
  - Disputed invoices are settled within 36 days (see Figure 8 on page 18), on average.
- c. Percentage of disputes received by the company that were lost (within two decimal places).
  - Of the 571 disputed invoices, 17.69 percent or 101 were lost (see Figure 1 on page 7).
- d. Percentage of revenue lost from disputes (within two decimal places).
  - The 101 lost disputes translated to a 4.67 percent loss (\$690,167), compared to the possible revenue if all invoices were collected. (see Problem Statement on page 7).
- e. The country where the company reached the highest losses from lost disputes (in USD).
  - Yellevate saw the highest revenue loss from disputed invoices in France (see Figure 3 on page 14).



# METHODOLOGY

## Data Sources, Tools, and Procedure

The recommendations and data analysis included in this report were derived from Yellevate's available data of the invoices it issued from 2020 to 2021. The available data points are described in the Glossary of Terms on page 10.

All the analysts in the team inspected the dataset individually to look for the issues in the dataset prior to analysis. No issues in the data were found and the provided dataset was used as is for the analysis. The issues investigated includes:

1. Spellings and misspellings for categorical data points
2. Logical errors in the dataset
  - a. entries where `disputed = FALSE` but `dispute_lost = TRUE`
  - b. entries where the given `days_to_settle` were incorrect if the number of days between `invoice_date` and `settled_date` were computed
3. Null or empty values
4. Duplicated entries
5. Data entry errors

Once the data was verified to be fit for analysis, all the analysts did independent work on the dataset for a week. The team then convened to discuss the findings. Data trends and insights were considered valid when at least two analysts arrived at the same numbers. Any disagreements were inspected and discussed by the team. Disagreements that arouse from differing assumptions and understandings of terms were ironed out.

Once no new insights, data trend and statistic have been discovered, the analysts proceeded to form independent recommendations. The same vetting process from the team was conducted, and proposals that received approval from at least two analysts were included in this report.

Only findings relevant to the recommendations were included in this report.

## Timeline

The project commenced on Sept. 5, 2022, and all the deliverables described have been submitted on Sept. 26, 2022 for Yellevate's review and perusal.

## Glossary of Terms

Yellevate provided a glossary for the data it provided for the analysis

Table 1. Dataset Glossary of Terms

| Field Name                                      | Data Type | Definition                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>country</b>                                  | text      | Country where the invoice was issued.                                                                                                                                                                                                                                                                                                                    |
| <b>customer_id</b>                              | text      | Unique identifier for each customer.                                                                                                                                                                                                                                                                                                                     |
| <b>invoice_number</b>                           | numeric   | Unique identifier for each invoice issued.                                                                                                                                                                                                                                                                                                               |
| <b>invoice_date</b>                             | date      | Date when invoice was generated.                                                                                                                                                                                                                                                                                                                         |
| <b>due_date</b>                                 | date      | Date when payment for invoice is due.                                                                                                                                                                                                                                                                                                                    |
| <b>invoice_amount_usd</b>                       | numeric   | Dollar value of invoice; how much the customer needs to pay.                                                                                                                                                                                                                                                                                             |
| <b>disputed</b>                                 | boolean   | "1" means the customer disputed the invoice; "0" means they did not.                                                                                                                                                                                                                                                                                     |
| <b>disputed_lost</b>                            | boolean   | "1" means Yellevate lost the dispute and the dispute was resolved in favor of the customer, and the customer does not have to pay the invoice; "0" means the customer did not win the dispute, and they are legally required to pay the full invoice amount, it either means that Yellevate won the dispute, or there was no dispute in the first place. |
| <b>settled_date</b>                             | date      | Date when either the customer paid the invoice, or if the dispute was resolved in favor of the customer, the decision that the customer does not have to pay was finalized.                                                                                                                                                                              |
| <b>days_to_settle</b> or <b>processing time</b> | numeric   | Number of days from when the invoice was generated to when it was settled.                                                                                                                                                                                                                                                                               |
| <b>days_late</b>                                | date      | Number of days from when the invoice was due to when it was settled, 0 when <code>due_date &gt; settled_date</code> and thus the invoice was settled on time.                                                                                                                                                                                            |

Additionally, the team has used several terms that may need clarifications

Table 2. Glossary of Additional Terms Used in This Report

| Term                     | Definition                                                                                                                                                                                                                                                 |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>chronic disputers</b> | The term refers to clients who have filed invoice disputes more than the average Yellevate client. Throughout this report, chronic disputers refers clients which has disputed eight or more invoices, regardless whether those disputes were won or lost. |
| <b>lost revenue</b>      | When used in this report, the terms refer to the revenue Yellevate lost from invoices that were not collected due to a lost dispute.                                                                                                                       |
| <b>revenue loss</b>      |                                                                                                                                                                                                                                                            |
| <b>lost dispute</b>      | Refers to disputes that resulted to a payment opt out by the client.                                                                                                                                                                                       |



# FINDINGS

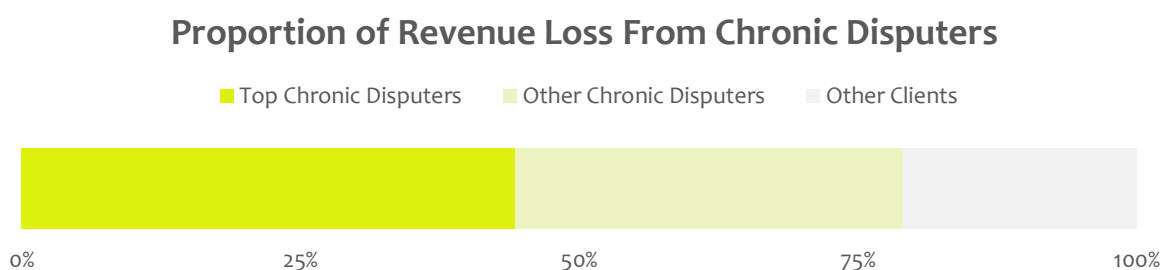
## Chronic Disputers Forms the Bulk of the Problem

In total, Yellevate has lost \$690 thousand from all disputes where the clients had the option not to pay (lost dispute) over the fiscal years 2020 and 2021. This accounts for roughly 4.67 percent of what could have been the company's annual revenue if all invoices were collected.

Of Yellevate's total of 100 clients for the covered years, the repeated disputes filed by five clients already accounts for \$305,589 or 44.48 percent of the company's losses. These clients are enumerated on Recommendation 1 (see page 19). These five represents the extreme of the "chronic disputers."

On average, the typical Yellevate client has disputed an invoice seven to eight times over 2020 and 2021. These chronic disputers are those that have disputed at least eight invoices. There are 22 other clients that fits this profile. The total would be 27 if we include the five mentioned above. The repeated disputes of these other 22 chronic disputers have cost the business roughly \$240,241 or 34.61 percent of the company's losses from disputes.

Figure 2. Proportion of Revenue Loss from Chronic Disputers

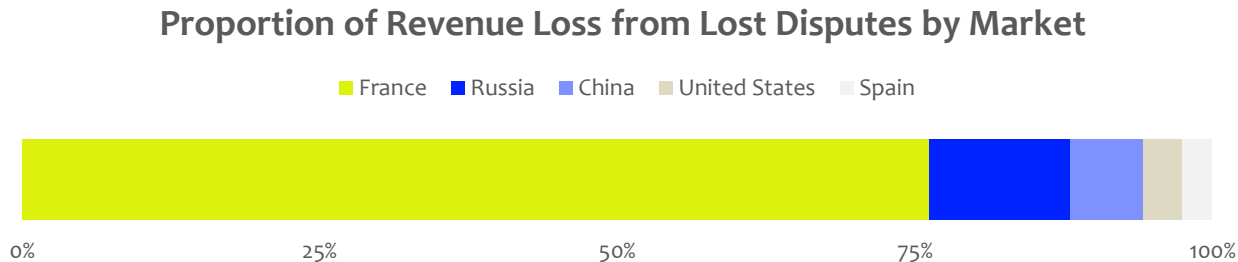


Therefore, by simply dealing with five percent of our clients — the five top chronic disputers — Yellevate can already solve almost half of its issues. Meanwhile, dealing with the other 22 chronic disputers will also help the company focus its resources on where it can make the most impact.

Most Chronic Disputers and Revenue Loss Is in France

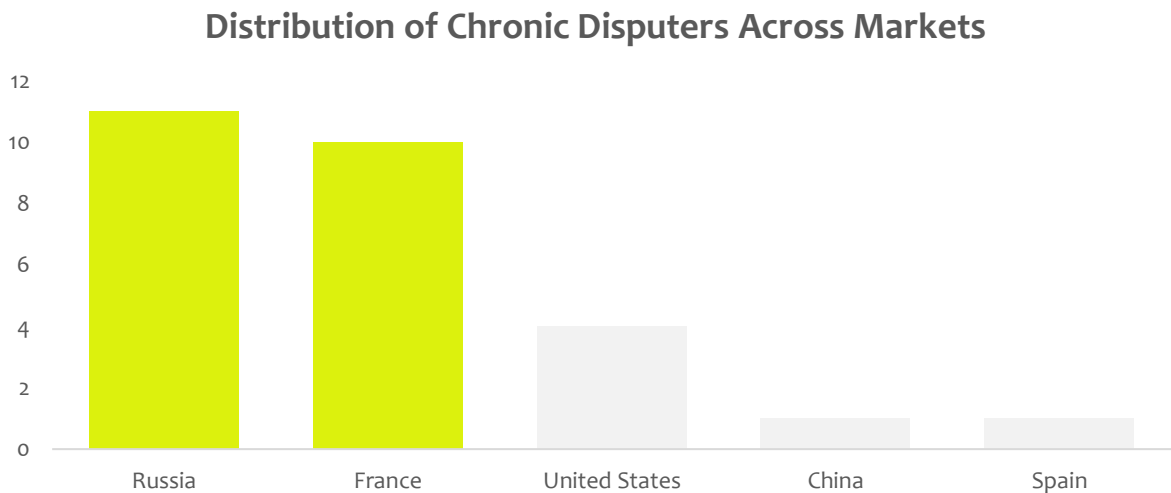
The recommendation to focus the first response to the problem in France is also founded on the fact that many of the chronic disputers (see Figure 4) are from France, and it is also the market where the business has lost the most revenue.

Figure 3. Proportion of Revenue Loss from Lost Disputes by Market



Although the Russian market has the most chronic disputers (see Figure 4), the country only accounts for 11.77 percent of the losses. That’s relatively smaller to the revenue loss from disputes we suffered from France (76.25%). Yellevate may want to focus on this country, Russia, only after implementing changes to the French market. After, effort can then be diverted to the problem disputers in Russia, China, U.S., and Spain — in that order (see Figure 3).

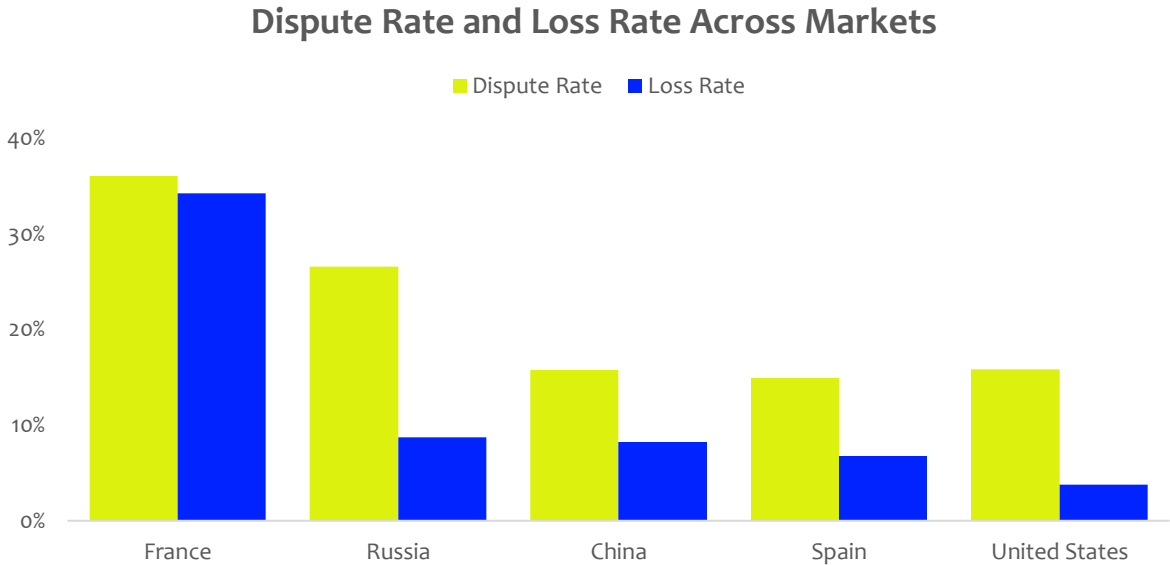
Figure 4. Distribution of Chronic Disputers Across Markets



When dealing with the other markets, Yellevate may want to implement blanket measures that may work across markets. Special

attention and resources thrown at these countries may not be warranted, given that the problem may not be as “big” in these areas. Recommendation 3 (see page 19) offers some ideas on what blanket measures may be implemented. However, further assessment of exact measures to be implemented may still be required.

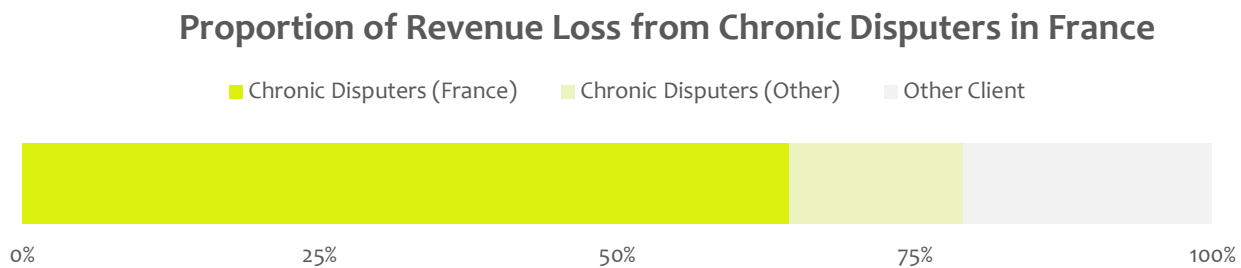
Figure 5. Dispute Rate and Loss Rate Across Markets



The percentage of disputes (dispute rate) and losses (loss rate) also points to France as a market of concern. The dispute rate offers us a picture of how often, on average, do we encounter disputes in a particular market. Meanwhile, the loss rate offers us a single number to assess how often do we lose these disputes. France has the largest disputed rate (36.04%). Meanwhile, of all those disputes — we lost 34.23 percent of them in France, the highest among all the markets.

If the focusing on the entire French market may still be a challenge, Yellevate can further prioritize. The most compelling group to target first when solving this issue are the 10 chronic disputers from France. These 10 already include the “top chronic disputers” described in Chronic Disputers Forms the Bulk of the Problem (see page 13) and Recommendation 1 (see page 19). They account for 64.40 percent of all the revenue Yellevate lost from disputes: dealing with them deals with half of the entire problem (see Figure 6).

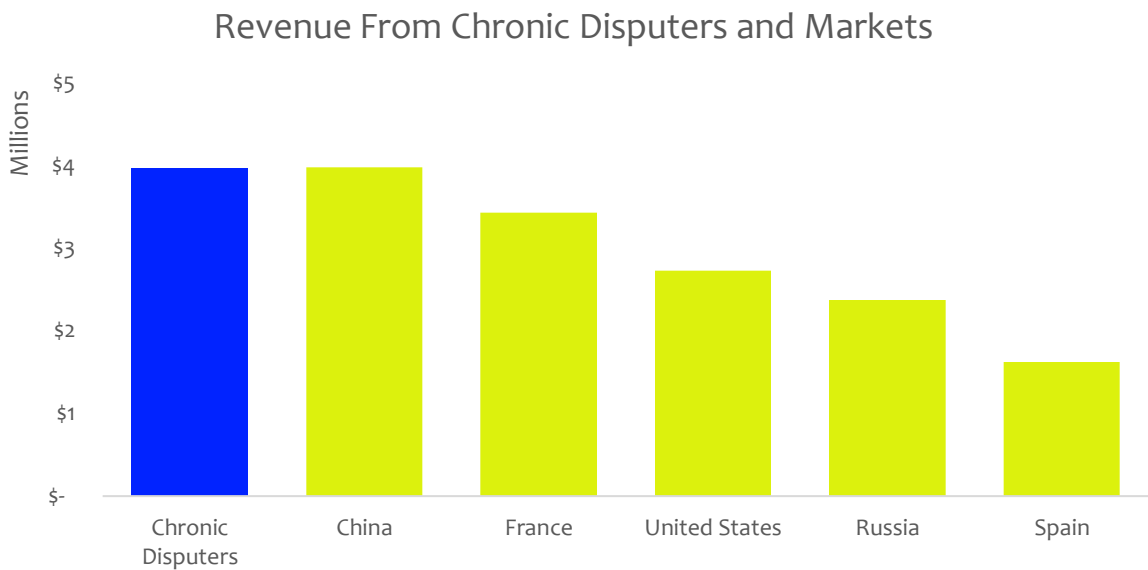
Figure 6. Proportion of Revenue Loss from Chronic Disputers in France



### Chronic Disputers Are a Large Source of Revenue

Yellevate’s revenue primarily comes from its Chinese and French clientele. The two countries combined accounts for half of the company’s total revenue with China accounting for \$3.9 million and France \$3.4 million (see Figure 7). That alone warrants that attention be given to France first as one of the largest markets where we are seeing the largest revenue loss from disputes.

Figure 7. Revenue From Chronic Disputers and Markets



If France is as crucial to the business as its revenue share suggests, it would be prudent to invest in good legal representation in the country to review the contracts of any loopholes that the French clientele are using to circumvent paying for the company’s services.



Further, this means that “letting go” of the chronic disputers may not be a wise course of action. While the French market accounts half of the company’s 4.67 percent revenue loss problem, about a quarter of Yellevate’s revenue (24.26%) comes from this market. We are, in fact, earning more from these chronic disputers than we are losing. The 27 chronic disputers, combined, brought in \$3.9 million over 2020 and 2021. That’s just as big as our revenue from our largest market (see Figure 7).

## Processing Time of Invoices

On average, invoices are paid within 26 days of its issuance with majority of the invoices being settled between 18 and 35 days (see Figure 8). This processing time is longer for disputed invoices which are settled by clients, on average, within 36 days — six days beyond Yellevate’s standard policy of 30-day payment term. Majority of the disputed invoices are settled between 28 and 44 days. The longer settlement time for disputed invoices is an expected characteristic of these transactions since the time allotted for the dispute process is reflected in the average processing time of disputed invoice. This added time is expected to not be characteristic of non-disputed invoices.

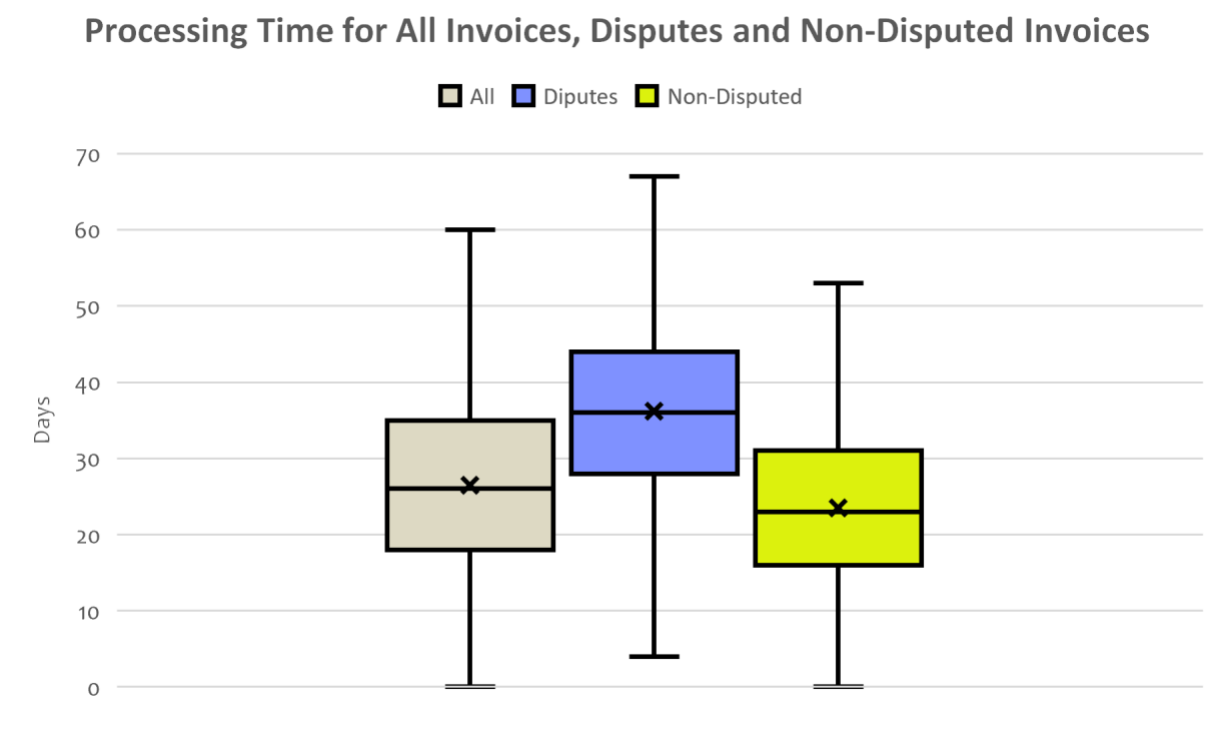
Data shows that roughly 25 percent of non-disputed invoices are paid late. While roughly 75 percent of the non-disputed invoices are settled on time, the remaining are paid between 31 days and 53 days.

If Yellevate could resolve problems on disputed invoices combined with the implementation of the above stated recommendations — particularly Recommendation 4 (see page 19) — the average time of processing invoices would be significantly reduced.

When offering incentives for early payment, Yellevate may want to offer discounts for payments made within 14 days of an invoice’s issuance. Majority of the non-disputed invoices are settled around 16 to 31 days with only roughly a quarter of invoices being settled earlier. This means that the company should try to “pull” the processing time down and 14 days seems to be a good number to target as it is just below the lower range of the business’ typical invoice processing time (see Figure 8). An early payment discount can incentivize clients who do not have issues with the company’s services to pay early. Invoices that are settled earlier helps the business by improving cash

flow. Further, the faster an invoice the less opportunities there are for a client to “change their mind” and start an invoice dispute process.

Figure 8. Processing Time for All Invoices, Disputes and Non-Disputed Invoices



# RECOMMENDATIONS

In summary, this team of analysts recommends the following courses of action to help limit Yellevate's further losses from invoice disputes:

1. Investigate the following customers closer. Their repeated disputes account for about half of the business' lost revenue in 2020–2021. Management may need to decide whether to keep them as clients or eventually let them go.
  - a. 3448-OWJOT
  - b. 9725-EZTEJ
  - c. 4632-QZOKX
  - d. 7600-OISKG
  - e. 9771-QLGZ
2. Invest in good legal representation in France.
  - a. Check the consumer laws in France. There might be inherent bias in favor of customers and find out what legal leverage of which the company can take advantage.
3. Instate a process that discourages invoice disputes, makes it harder to file, and more difficult to win.
  - a. Bill clients on a “per project milestone” basis instead of at the end of a project.
  - b. Require upfront payments prior to the start of the project.
  - c. Include a contract clause that requires all disputes to be settled through arbitration by a third party to be designated by Yellevate.
  - d. Include a contract clause that, if a dispute is settled in favor of Yellevate, the client will have to reimburse the company for what is spent on administrative and representation fees related to the dispute.
4. Incentivize “good client behavior,” and impose penalties for unwanted behavior.
  - a. Offer rewards (i.e. discounts or other perks) for invoices paid early, particularly those settled within 14 days of the invoice date.
  - b. Add a contract penalty clause for when an account fails to pay the full within Yellevate's set standard payment terms of 30 days.

